Let $x_1, \cdots, x_n \overset{i.i.d.}{\sim} P$ be square integrable stochastic process where P is supported on a compact set $\mathcal{T} \subset \mathbb{R}$. Let $y_i = \langle x_i, \beta_0 \rangle + \varepsilon_i$ (w.l.o.g. suppose x is centered and we omit the intercept here). Let $\boldsymbol{K}$ be a known reproducing kernel and let $\mathcal{H}(\boldsymbol{K})$ be the corresponding RKHS. Assume that $\beta_0 \in \mathcal{H}(\boldsymbol{K})$. Let $\|\cdot\|_{\psi_2}$ denote the sub-gaussian norm of a random variable. Suppose $L_{\boldsymbol{K}^{1/2}} x$ is sub-gaussian (has bounded sub-gaussian norm).

Let $\boldsymbol{C}_0$ be the covariance function of x defined as follows:

$$\boldsymbol{C}_0(s,t) = \mathbb{E}[(x(s) - \mathbb{E}[x(s)])(x(t) - \mathbb{E}[x(t)])].$$

Suppose that the eigenvalues s_k of the operaotr $\boldsymbol{T}_0 = L_{\boldsymbol{K}^{1/2}\boldsymbol{C}_0\boldsymbol{K}^{1/2}}$ have $s_k \asymp k^{-2r}$. We have the subgaussian norm $\sigma_x \asymp s_1$. (We shall assume that $r > 1/2$ such that the trace of operator is bounded). Let also $\boldsymbol{T}_\lambda = \boldsymbol{T}_0 + \lambda \mathbf{1}$. Denote $\widehat{\boldsymbol{T}}_n$ to be

$$\widehat{\boldsymbol{T}}_n = L_{\boldsymbol{K}^{1/2}\boldsymbol{V}_n\boldsymbol{K}^{1/2}}$$

where $\boldsymbol{V}_n(s,t) = \frac{1}{n}\sum_{i=1}^n (x_i(s) - \bar{x}(s))(x_i(t) - \bar{x}(t))$ and let $\widehat{\boldsymbol{T}}_\lambda = \widehat{\boldsymbol{T}}_n + \lambda \mathbf{1}$.

Throughout we omit poly-logarithmic terms in $\widetilde{O}(\cdot)$ and $\lesssim$ notations. When we say "with high probability", it means with probability $1 - O(T^{-4})$ or higher degrees.

The objective is to construct UCB $\Delta(\cdot)$ so that it satisfies the following properties:

1. With high probability $|x^\top(\widehat{\beta} - \beta_0)| \leq \Delta(x)$ for all $x \in \mathcal{X}$, where $\mathcal{X}$ contains T vectors in the support of P;
2. $\mathbb{E}_P[\Delta(x)] \leq \Psi$ where Ψ is on the order of $\widetilde{O}(1/\sqrt{n})$.

Note that we don't require $\Delta(x)$ to be valid *uniformly* on the support of P, but it has to be valid pointwise with high probability so that a union bound of T vectors won't hurt.

0.1 Direct analysis of ridge regression when T is known

Define the ridge regressor of $\widetilde{\beta}_0 = (L_{\boldsymbol{K}^{1/2}})^{-1}\beta_0$ to be

$$\widehat{\beta}_\Lambda = \widehat{\boldsymbol{T}}_\Lambda^{-1}(\widehat{\boldsymbol{T}}_n\widetilde{\beta}_0 + g_\epsilon)$$

where

$$g_\epsilon = \frac{1}{n}\sum_{i=1}^n \varepsilon_i L_{\boldsymbol{K}^{1/2}} x_i.$$

Suppose that $\widetilde{x}$ is a gaussian process and $\widetilde{x} \sim \mathcal{N}(0, \Sigma)$, where Σ is the covariance operator. Than we have with high probability

$$\langle \widetilde{x}, \widetilde{\beta}_0 - \widehat{\beta}_{\boldsymbol{\Lambda}} \rangle \leq$$

For any $\widetilde{x}$ we have

$$\langle \widetilde{x}, \widetilde{\beta}_0 - \widehat{\beta}_{\boldsymbol{\Lambda}} \rangle = \langle \widetilde{x}, \widetilde{\beta}_0 - \widetilde{\beta}_{\boldsymbol{\Lambda}} \rangle + \langle \widetilde{x}, \widetilde{\beta}_{\boldsymbol{\Lambda}} - \widehat{\beta}_{\boldsymbol{\Lambda}} \rangle,$$

where

$$\widetilde{\beta}_{\boldsymbol{\Lambda}} = \boldsymbol{T}_{\boldsymbol{\Lambda}}^{-1}\boldsymbol{T}\widetilde{\beta}_0.$$

Analysis of the population bias term. We have

$$\langle \widetilde{x}, \widetilde{\beta}_0 - \widehat{\beta}_{\boldsymbol{\Lambda}} \rangle = \langle \widetilde{x}, -\boldsymbol{\Lambda}\boldsymbol{T}_{\boldsymbol{\Lambda}}^{-1}\widetilde{\beta}_0 \rangle = \sum_{j=1}^\infty \widetilde{\theta}_j \frac{\alpha_j \lambda_j}{\lambda_j + s_j}.$$

Analysis of the bias term. We have

$$\langle L_{\boldsymbol{K}^{1/2}}x, \lambda\widehat{\boldsymbol{T}}_\lambda^{-1}f_0\rangle = \lambda\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}f_0\rangle = \lambda\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}\widehat{f}_\lambda\rangle + \lambda\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}(f_0-\widehat{f}_\lambda)\rangle. \quad (1)$$

For the RHS, the former term is explicit, while we do not know the second term. Thus we can write

$$\begin{aligned}\lambda\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}(f_0-\widehat{f}_\lambda)\rangle &\le \lambda\left|\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-3/2}L_{\boldsymbol{K}^{1/2}}x\rangle\right| \left\|\widehat{\boldsymbol{T}}_\lambda^{1/2}(f_0-\widehat{f}_\lambda)\right\|_{\mathcal{L}_2} \\ &\preceq \lambda\left|\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-3/2}L_{\boldsymbol{K}^{1/2}}x\rangle\right| \left\|\boldsymbol{T}_\lambda^{1/2}(f_0-\widehat{f}_\lambda)\right\|_{\mathcal{L}_2},\end{aligned}$$

given $\widehat{\boldsymbol{T}}_\lambda \succeq (1/2)\boldsymbol{T}_\lambda$. (To obtain this, we need to use Berstein that consider the effective dimension like [1]). We further have

$$\left\|\boldsymbol{T}_\lambda^{1/2}(f_0-\widehat{f}_\lambda)\right\|_{\mathcal{L}_2} \le \sqrt{\lambda} + \left\|\boldsymbol{T}_0^{1/2}(f_0-\widehat{f}_\lambda)\right\|_{\mathcal{L}_2}$$

and

$$\left\|\boldsymbol{T}_0^{1/2}(f_0-\widehat{f}_\lambda)\right\|_{\mathcal{L}_2} \preceq \sqrt{\lambda} + \left\|\boldsymbol{T}_0^{1/2}\boldsymbol{T}_\lambda^{-1/2}g_\varepsilon\right\|_{\mathcal{L}_2} \preceq \sqrt{\lambda} + \sqrt{\frac{1}{n}}\sqrt{\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}L_{\boldsymbol{K}^{1/2}}x\rangle},$$

where the first term on the RHS is the deterministic bias term and the second term is the variance term. Putting everything together we have that the point-wise bias in (1) can be upper bounded by

$$\begin{aligned}&\langle L_{\boldsymbol{K}^{1/2}}x, \lambda\widehat{\boldsymbol{T}}_\lambda^{-1}f_0\rangle \\ &\preceq \lambda\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}\widehat{f}_\lambda\rangle + \lambda\left|\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-3/2}L_{\boldsymbol{K}^{1/2}}x\rangle\right| \left(\sqrt{\lambda} + \sqrt{\frac{1}{n}}\sqrt{\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}L_{\boldsymbol{K}^{1/2}}x\rangle}\right) \\ &:= \Delta_b^1(x) + \Delta_b^2(x) := \Delta_b(x).\end{aligned}$$

Analysis of the variance term. We have the variance term

$$v = \langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}g_\epsilon\rangle = \frac{1}{n}\sum_{i=1}^n \varepsilon_i\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}L_{\boldsymbol{K}^{1/2}}x_i\rangle.$$

Since ε_i are i.i.d. σ^2 subgaussian, we have

$$v \preceq \sqrt{\frac{1}{n}}\sqrt{\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-2}\widehat{\boldsymbol{T}}_n L_{\boldsymbol{K}^{1/2}}x\rangle} \le \sqrt{\frac{1}{n}}\sqrt{\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}L_{\boldsymbol{K}^{1/2}}x\rangle} := \Delta_v(x).$$

Putting bias and variance together. We can now define

$$\Delta(x) := \Delta_b(x) + \Delta_v(x)$$

and bound $\mathbb{E}_P[\Delta(x)]$. For the bias term, we have

$$\mathbb{E}\left[\Delta_b^1(x)\right] \preceq \lambda\sqrt{\mathbb{E}\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}\widehat{f}_\lambda\rangle^2\right]} \preceq \lambda\sqrt{\mathbb{E}\left[\langle L_{\boldsymbol{K}^{1/2}}x, \boldsymbol{T}_\lambda^{-1}\widehat{f}_\lambda\rangle^2\right]} \preceq \lambda^{1/2}$$

and

$$\mathbb{E}_P\left[\Delta_b^2(x)\right] \preceq \lambda\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-3/2}L_{\boldsymbol{K}^{1/2}}x\rangle^2\right]}\left(\sqrt{\lambda} + \sqrt{\frac{1}{n}}\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}L_{\boldsymbol{K}^{1/2}}x\rangle\right]}\right).$$

For the first expectation on the RHS, we have

$$\lambda\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-3/2} L_{\boldsymbol{K}^{1/2}}x\rangle^2\right]} \preceq \lambda\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \boldsymbol{T}_\lambda^{-3/2}\rangle^2\right]} = \sqrt{\sum_{k=1}^{\infty}\frac{s_k\lambda^2}{(\lambda+s_k)^3}}.$$

which yields

$$\begin{aligned}\mathbb{E}_P\left[\Delta_b^2(x)\right] &\preceq \sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \boldsymbol{T}_\lambda^{-1} L_{\boldsymbol{K}^{1/2}}x\rangle^2\right]}\left(\sqrt{\lambda}+\sqrt{\frac{1}{n}}\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1} L_{\boldsymbol{K}^{1/2}}x\rangle\right]}\right)\\ &\preceq \sqrt{\lambda}+\sqrt{\frac{1}{n}}\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1} L_{\boldsymbol{K}^{1/2}}x\rangle\right]},\end{aligned}$$

since $\sqrt{\mathbb{E}_P\left[\langle L_{\boldsymbol{K}^{1/2}}x, \boldsymbol{T}_\lambda^{-1} L_{\boldsymbol{K}^{1/2}}x\rangle^2\right]} = o(1)$. We further have

$$\begin{aligned}\sqrt{\mathbb{E}\left[\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1} L_{\boldsymbol{K}^{1/2}}x\rangle\right]} &\preceq \sqrt{\mathbb{E}\left[\langle L_{\boldsymbol{K}^{1/2}}x, \boldsymbol{T}_\lambda^{-1} L_{\boldsymbol{K}^{1/2}}x\rangle\right]}\\ &= \sqrt{\sum_{k=1}^{\infty}\frac{s_k}{\lambda+s_k}},\end{aligned}$$

where s_k are the eigenvalues of $\boldsymbol{T}_0$ given that $\widehat{\boldsymbol{T}}_\lambda \succeq (1/2)\boldsymbol{T}_\lambda$. Since $s_k \asymp k^{-2r}$, we have

$$\sqrt{\sum_{k=1}^{\infty}\frac{s_k}{\lambda+s_k}} \asymp \exp(\log\lambda/(-2r)).$$

For the variance term, we similarly have

$$\mathbb{E}_P\left[\Delta_v(x)\right] \leq \sqrt{\frac{1}{n}\exp(\log\lambda/(-2r))}.$$

Putting everything together, we have

$$\mathbb{E}_P\left[\Delta_b(x)+\Delta_v(x)\right] \preceq \mathbb{E}_P\left[\Delta_b^1(x)\right]+\mathbb{E}_P\left[\Delta_b^2(x)\right]+\mathbb{E}_P\left[\Delta_v(x)\right] \preceq \sqrt{\lambda}+\sqrt{\frac{1}{n}\exp(\log\lambda/(-2r))}.$$

By choosing $\lambda \asymp n^{-2r/(2r+1)}$ we have

$$b+v \asymp n^{-r/(2r+1)},$$

which is minimax optimal.

References

[1] Stanislav Minsker. On some extensions of bernstein's inequality for self-adjoint operators. *arXiv preprint arXiv:1112.5448*, 2011.